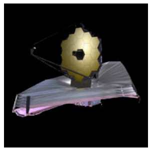

5370 - Eyes on the Stars: A JWST Population Survey of Exoplanet Host Star Heterogeneities and Spectral Contributions to Transits

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Early JWST results have shown both the observatory’s promise for revealing the properties of exoplanet atmospheres and the challenge that stellar spots and faculae pose to precise transmission spectroscopy. Without a reliable method to account for this stellar spectral contamination, stellar photospheric heterogeneity may limit JWST's scientific legacy in unveiling the atmospheres of the most exciting exoplanets. Fortunately, the same

JWST datasets provide exquisite out-of-transit stellar spectra that can be leveraged to mitigate this effect, though most observations have yet to be fully exploited. Here we propose to thoroughly tap this priceless database via a large, uniform analysis of 36 NIRISS/SOSS and NIRSpec/PRISM time-series observations of 21 exoplanet hosts. Our analysis will reveal the photospheric properties of important hosts, provide insights into limitations of stellar models across spectral types, and constrain stellar variability and spectral contamination in JWST transits. As Legacy Data Products, we will produce a JWST Legacy Library of time-dependent exoplanet host star spectra, constraints on their photospheric spectral components and coverages, and transmission spectra with fully quantified uncertainties accounting for stellar spectral contributions. Our Legacy Data Products will also include open-source Python libraries we develop for the data reduction, stellar heterogeneity analysis, and transmission spectra corrections. The results of this program will pave the way for robust constraints on stellar photospheres from JWST, unlocking the observatory's full potential for characterizing transiting worlds.

OBSERVING DESCRIPTION